

Geometry, Perimeter & Area — Level 2 (Compound Shapes)

A dedicated harder follow-on to Week 7, for students who've qualified Level 1. Level 1 asks for the perimeter or area of a single rectangle, square, or triangle; Level 2 combines shapes — paths, frames, cut corners, and figures built from several pieces, plus basic quadrilateral angle sums. All questions are solved using only the plain rectangle/square formulas and the angle-sum facts already in the syllabus — no new formulas needed, just combining them. 15 questions, 4-option MCQ, untimed.

Q1. A rectangular garden is 30 m long and 22 m wide. A path of uniform width 3 m is built all around the inside edge of the garden (so the path runs just inside the boundary). Find the area of the path. A) 256 sq m B) 266 sq m C) 276 sq m D) 286 sq m

Q2. A rectangular park is 40 m long and 28 m wide. A path of uniform width 3 m is built all around the OUTSIDE of the park. Find the area of the path. A) 424 sq m B) 434 sq m C) 444 sq m D) 454 sq m

Q3. A rectangular photo frame has outer dimensions 50 cm by 35 cm. The wooden border is 4 cm wide all the way around. Find the area of the wooden border itself (not the photo it holds). A) 596 sq cm B) 606 sq cm C) 616 sq cm D) 626 sq cm

Q4. An L-shaped room is made by joining two rectangular sections: one 7 m by 5 m, and another 4 m by 3 m. Find the total floor area of the room. A) 43 sq m B) 45 sq m C) 47 sq m D) 49 sq m

Q5. A square of side 4 cm is cut out from one corner of a rectangular sheet of paper measuring 14 cm by 9 cm. Find the area of the paper that remains. A) 104 sq cm B) 107 sq cm C) 110 sq cm D) 113 sq cm

Q6. If both the length and the breadth of a rectangle are made 3 times as long, how many times bigger does its perimeter become? A) 2 times B) 3 times C) 6 times D) 9 times

Q7. If the side of a square is made 4 times as long, how many times bigger does its area become? A) 4 times B) 8 times C) 12 times D) 16 times

Q8. Six identical square tiles, each of side 5 cm, are placed in a row, touching edge to edge, to form one long rectangle. Find the perimeter of this rectangle. A) 64 cm B) 67 cm C) 70 cm D) 73 cm

Q9. Eight identical rectangular tiles, each 6 cm by 4 cm, are placed together with no gaps or overlaps to form one bigger rectangle. Find the total area of the bigger rectangle. A) 182 sq cm B) 187 sq cm C) 192 sq cm D) 197 sq cm

Q10. A rectangular field is 50 m long and 32 m wide. A square plot of side 8 m is marked out inside the field. Find the area of the field that is NOT covered by the square plot. A) 1,516 sq m B) 1,526 sq m C) 1,536 sq m D) 1,546 sq m

- Q11.** A rectangular cardboard sheet is 14 cm by 9 cm. A 3 cm by 3 cm square notch is cut out of one corner (the cut edges run parallel to the original sides), leaving an L-shaped piece. Find the perimeter of the remaining L-shaped piece. A) 40 cm B) 43 cm C) 46 cm D) 49 cm
- Q12.** A rectangle is formed by placing two identical squares side by side, touching along one full edge. How many lines of symmetry does this rectangle have? A) 1 B) 2 C) 3 D) 4
- Q13.** Three angles of a quadrilateral are 80° , 95° , and 110° . Find the fourth angle. A) 65° B) 70° C) 75° D) 80°
- Q14.** A figure drawn on squared paper is an 8×6 grid of unit squares, with a 3×2 block of unit squares removed from one corner. Find the area of the figure, in square units. A) 36 B) 39 C) 42 D) 45
- Q15.** Two square tiles, each of area 64 sq cm, are placed side by side (touching along one full edge) to form a rectangle. Find the perimeter of this rectangle. A) 40 cm B) 44 cm C) 48 cm D) 52 cm
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Answer key: `answer-key.md`
